Applied Engineering & Robotics (8th Grade) — Syllabus

School Year: 2026–2027
Instructor: Matt Bombich
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Room: 70
Office Hours: Monday–Thursday, 2:45–3:15 pm
Prerequisites: None — curiosity required.

Course Description

Applied Engineering & Robotics (8th Grade) is a one-semester, hands-on course where you experience engineering the way professionals do it. You will use the Engineering Design Process to design, model, and build parts for L.E.O. (Logic, Electronics, and Operations) — a real autonomous robot — using professional CAD software, 3D printers, and programming tools.

This course gives you a strong foundation in engineering thinking and technology skills before high school.

Course Objectives

By the end of the semester, you will be able to:

- Use the Engineering Design Process to define problems and propose solutions.
- Model 3D parts in Autodesk Fusion 360 and iterate based on caliper measurements.
- Program a Raspberry Pi Pico 2W in MicroPython for basic input/output and motor control.
- Safely operate a 3D printer, drill press, and hand tools with proper PPE.
- Reflect on your design decisions in a digital portfolio.

Materials & Tools

All materials and tools are provided. You do not need to purchase anything.

Category	Items
Electronics	Raspberry Pi Pico 2W, L.E.O. interface board, basic sensors
Software	Autodesk Fusion (CAD), Thonny IDE (MicroPython), Google Workspace
Fabrication	Bambu and Creality 3D printers, drill press, hand tools, calipers
Safety	Safety glasses, gloves (provided)
Resources	Class portfolio site, Google Classroom

Grading

Category	Weight

Portfolio Deliverables	50%
Quizzes	20%
Final Exam	20%
Safety, 5S & Participation	10%

Portfolio Deliverables (50%) are the core of your grade. Each major milestone — your Design Brief, CAD components, safety certifications, and programming assignments — is submitted through your digital portfolio. Each deliverable has a rubric visible before you submit. Point values reflect the complexity of the work.

Quizzes (20%) are given at the end of major units to assess understanding of content and skills covered.

Final Exam (20%) is a cumulative assessment given at the end of the semester.

Safety, 5S & Participation (10%) — You begin the semester with 50 points. Ten points are deducted per incident, including: not wearing required PPE, leaving tools or materials out after use, failing to participate in 5S cleanup, or other unsafe behaviors. Serious or repeated violations are referred to administration regardless of remaining points.

Unit & Pacing Overview

Unit	Title	Approx. Weeks	Key Skills
1	Engineering Design Process	1–2	EDP steps, design brief, problem definition
2	CAD & 3D Printing	3–8	Fusion 360 modeling, caliper measurement, iterative printing (C1–C2)
3	Shop Safety	9	Drill press, PPE, LOTO, SDS — certification before build phase
4	Programming & Robotics	10–16	MicroPython basics, motor control, IR sensors, simple autonomous behaviors
6*	Career Readiness	ongoing	Career exploration, paycheck analysis, portfolio reflection

Exact pacing is posted on the Class Hub. Weeks are approximate.

** Unit 5 (AI & Machine Learning) is not included in the 8th grade course. The one-semester format requires a condensed scope; Unit 5 is covered in the full-year high school course.*

Classroom Policies

Safety — Non-Negotiable
PPE is required whenever shop tools or electronics are in use. A safety violation results in a 10-point deduction from your Safety, 5S & Participation grade and removal from the work area. Repeated violations will be referred to administration.

Attendance
Lab time cannot be recreated from home. Excused absences receive full make-up opportunities.

Late Work
Portfolio deliverables are accepted up to 3 school days late with a 10% per day penalty. Talk to Mr. Bombich before the deadline if you need an extension.

Academic Integrity
AI tools may be used to understand concepts or get unstuck — but submitted work must reflect your own thinking. Copying is academic dishonesty and handled per school policy.

Accommodations
Accommodation plans are fully supported. Please discuss your needs privately and early — accommodations work best when we plan ahead together.

Contact & Resources

Portfolio	mbombich-robotics.github.io/CTE
Class Website	mbombich-robotics.github.io/CTE/hub.html — all assignments, rubrics, pacing, and announcements
Email	mbombich@vicksburgschools.org — replies within 24 hours on school days
Tutorial	Tuesdays & Thursdays — request your assignment by end of day Monday or Wednesday. Students with missing work are assigned automatically.